

Arcade Tournament

Challenge 1

Give a player an award for their score.

Build it

1. Ask for the player's name and store it in `name`.
2. Ask for a total score from 0 to 30. Use `int()` to store it in `total`.
3. Use `if`, `elif` and `else` to display `name` followed by "earned Gold" for 24 or more, "earned Silver" for 15 to 23, and "earned Bronze" for anything lower.

Test it

- "Sam" and 24 display "Sam earned Gold".
- 15 displays "Sam earned Silver" and 14 displays "Sam earned Bronze".

Talk about it

Why must the Gold check come before the Silver check?

Save as `challenge1.py`.

Arcade Tournament

Challenge 2

Add up three round scores before giving the award.

Build it

1. Store 0 in `total`.
2. Use a `for` loop and `range()` to play rounds 1 to 3. In each round, display "Round" and the round number, ask for a score from 0 to 10, and add it to `total`.
3. After the loop, display "Total:" followed by `total`, then the award from Challenge 1.

Test it

- 8, 8 and 8 display "Total: 24" and Gold.
- 5, 5 and 5 display "Total: 15" and Silver.

Talk about it

Why is `total` set to 0 before the loop and not inside it?

Save as `challenge2.py`.

Arcade Tournament

Challenge 3

Let the player choose how many rounds to play, up to five.

Build it

1. Replace the `for` loop with a `while` loop. Store 0 in `rounds` and `"yes"` in `again`.
2. Keep playing while `again` is `"yes"` and `rounds` is less than 5. In each round, add 1 to `rounds`, ask for a score and add it to `total`, then ask `"Another round? yes/no"`.
3. After the loop, display `"Rounds played:"` followed by `rounds`, then `"Total:"` with `total` and the award.

Test it

- 9, `"yes"`, 9, `"yes"`, 9, `"no"` display `"Rounds played: 3"`, `"Total: 27"` and Gold.
- Answering `"yes"` every time stops after round 5.

Talk about it

Why is a `while` loop a better fit than a `for` loop for this version?

Save as `challenge3.py`.